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Vertically Situated Multifunctional Exercise Machine

Abstract

An embodiment of a multifunctional exercise machine that operates in the vertical plane (**44**) whereby a carriage (**42**) with or without anchoring points and openings for attachments and other accessories is physically propelled along a wall-mounted rail (**20**). An alternate embodiment of a multifunctional exercise machine that operates in the vertical plane (**44**) whereby a carriage (**42**) with or without anchoring points and openings for attachments and other accessories is physically propelled on an apt frame or structure. Another alternate embodiment of a multifunctional exercise machine that operates in the vertical plane (**44**) whereby a carriage (**42**) that is physically propelled on an apt structure or frame is fitted with known or unknown kinetic power generating devices or pumping systems thereby being useful with accomplishing a wide array of tasks that may suit a multitude of needs beyond exercise.

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Background/Summary

FIELD OF INVENTION

[0001] The present disclosure relates to multifunctional exercise machines of the type in which a carriage is propelled along a rail.

BACKGROUND ART

[0002] Heretofore, multifunctional exercise machines (MEM) of the type in which a carriage is manually propelled along a track, frame, or rail (rail) have been a popular means for exercise. These particular types typically utilize a carriage that operates within some variable degree of the horizontal plane. MEM that utilize a manually propelled carriage that operates within some degree of the horizontal plane (HMEM) have proven to be a somewhat useful means to perform a quantity of exercise, but unfortunately exercise on these devices typically fails to simultaneously benefit the intrinsic stiffness of active muscles as well as the neuromuscular reflex responses. In addition, most HMEM commonly fail to effectively decrease the time needed to perform a given routine; can be somewhat precarious at adequately supporting the user during activity; and most require a dedicated floor space. These aforementioned failures of the current catalog of HMEM can hamstring the likelihood that they will continue to be considered worthwhile devices.

[0003] Some devices from the current catalog of HMEM utilize cables and pulleys to manually propel body weight that is situated atop of a carriage through a series of movements or activities. Other HMEM utilize a carriage in conjunction with a frame or bar to push or pull weighted plates or cabled stacks; typically to perform an isolated upper or lower extremity activity.

[0004] There are some obvious disadvantages to HMEM. Firstly, the functional orientation of their carriage fails to provide the benefits of performing exercise in a true weight bearing position (vertical plane) thereby failing to adequately elicit neuromuscular reflex responses. Neuromuscular reflex responses (stabilizers) work in concert with the voluntary activation patterns in the torso musculature to provide spinal stability. These stabilizers function predominately in the vertical plane; they are a vital component of maintaining balance and coordination in relation to gravity. Stabilizers have been shown to work in concert with the active muscles of the torso, contributing to over 40% of total spinal stability; intrinsic muscle stiffness of active muscles alone is insufficient for optimal stability.

[0005] In addition to the disadvantages of the functional orientation of their carriage, most HMEM lack anchoring points and/or openings for attachments and other accessories that move in synchrony with said carriage providing for the consistent alignment of muscle fibers throughout a given activity (kinetic anchors); the ones that do provide said anchoring points typically utilize them exclusively as a means to manually propel their said carriage.

[0006] Also, to their detriment, HMEM have a footprint; most requiring some degree of floor space in a home, office, spa, gym, business, clinic or facility (facility) that can oftentimes be unavailable or unaffordable.

[0007] Still, there are other MEM such as cable-pulley devices, racks, and plate machines that are utilized to exercise in an upright position. These apparatuses typically lack a physically propelled carriage, do not have kinetic anchors, and often fail to support the axial skeleton through a full range of motion. In addition, they typically require a degree of dedicated floor space.

[0008] Worth mentioning, inflatable devices (IDT) have been utilized to perform activity in the vertical plane; these IDT lack both a carriage and kinetic anchors, can be somewhat precarious to use, and most require stowing.

[0009] In summation, the current catalog of HMEM and IDT typically fail to provide attributes that can make them worthwhile devices.

BRIEF SUMMARY OF THE INVENTION

[0010] In light of these shortcomings, what is needed is a multifunctional exercise machine whereby a carriage is physically propelled along a wall-mounted rail situated in the vertical plane (VMEM). Exercising with a VMEM can simultaneously benefit the intrinsic stiffness of active

muscles as well as the neuromuscular reflex responses (functional efficiency), and as an added result decrease the time needed to perform a given routine. In addition, a VMEM adequately supports the user during activity (i.e. provides safety) and requires a zero footprint.

[0011] Performing exercise against a VMEM can simultaneously activate lower extremity and torso muscle groups as well as stabilizers while providing axial support for the user throughout a full range of weight-bearing activity. This type of exercise can have many benefits. For instance, exercise against a VMEM supports optimal posture which alone can maximize respiratory capacity thereby increasing tissue oxygenation (studies have shown that slight variations in posture, more specifically torso flexion and forward head carriage especially during physical exertion, can noticeably decrease respiratory volume when measured with spirometry); exercise in the vertical plane can have the potential to significantly optimize therapeutic outcomes associated with bone and joint health (studies have shown that exercise that incorporates gravity as resistance is superior for maintaining spinal and lower extremity bone density as well as optimizing connective tissue strength and remodeling, both of which can contribute to improving the overall health and resilience of weight bearing joints); by supporting the axial skeleton against the carriage through the full range of motion of an exercise, a VMEM can resultantly optimize joint stability thereby increasing the safety of performing both low-intensity (Type 1 muscle fiber activation) and high-intensity (Type 2 muscle fiber activation) exercise; performing exercise in the vertical plane optimizes the mechanical benefits that can be obtained by activating the reflexive neuromuscular components that contribute to spinal stability (studies have shown that individuals with disturbed neuromuscular reflex responses may be more susceptible to spinal instability events).

[0012] It's also important to consider that most daily activities are complex movements; rarely being isolated tasks. Daily activities typically require coordinated activity between the upper extremity, lower extremity and torso muscle groups as well as stabilizers. To further optimize the functional efficiency of a VMEM, kinetic anchors should be available; the benefit being that upper extremity and/or torso muscle groups can be trained simultaneously with lower extremity and/or torso muscle groups along with stabilizers (synchronous activity).

[0013] It's logical to use kinetic anchors in conjunction with a VMEM. Resistance bands, tubing or attachments (attachments) or other accessories that are connected by kinetic anchors to the carriage of a VMEM can be utilized to perform a host of upper extremity and/or torso activities. Kinetic anchors located at various places on a VMEM move in synchrony with the carriage. This synchronous movement affords a consistent biomechanical orientation throughout the performance of a desired upper extremity and/or torso activity. For example, an upper extremity activity such as a chest or shoulder press can be performed with attachments connected with kinetic anchors while simultaneously performing a squat or calf raise that utilizes the sliding action of the vertically oriented carriage to guide a lower extremity press-up from the floor. The carriage functions to support the user throughout an activity's full range of motion and can be situated at numerous locations along the axial skeleton to accommodate the desired alignment of a chosen upper extremity activity; the alignment of said upper extremity activity remains constant throughout the movement because kinetic anchors move in synchrony with the carriage. In addition to the simultaneous activation of both an upper and lower extremity muscle group as discussed in this example, the weight-bearing orientation of a user performing exercise in the vertical plane elicits a reflexive neuromuscular response, i.e. activates stabilizers.

[0014] Training and/or rehabilitation routines done utilizing synchronous activity can enhance the way our bodies function while performing daily activities; these routines can improve neurological coordination and boost the efficiency of gait; as well as strengthen those muscle groups that can help counteract the negative effects of sedentary habituations. Activity performed against a VMEM with kinetic anchors further optimizes functional efficiency by providing for synchronous activity while the carriage simultaneously affords a significant measure of safety.

[0015] Kinetic anchors can also be utilized to link to stationary anchor points that are adjacent to

the carriage or situated at various locations on the rail (static anchors). Linking kinetic anchors to static anchors with attachments can provide the option for resistance to the upward/downward glide of the carriage along a vertically oriented rail. Static anchors can also allow for a multitude of synchronous activity to be performed utilizing attachments and other accessories. In addition, static anchors can be utilized to perform a wide array of traditional exercise that doesn't require synchronous activity. Static anchors are not novel, but can nonetheless provide a host of functional features that increase the usefulness of a VMEM.

[0016] Time conservation is an another apparent benefit of a VMEM; simultaneously training upper extremity, lower extremity and/or torso muscle groups along with stabilizers can optimize an exercise routine thereby reducing the duration of time typically needed to individually train those same muscle groups.

[0017] As previously discussed, worthwhile exercise equipment should also consider the physical footprint required to accommodate a device. A VMEM, even with kinetic anchors, demonstrates a true zero footprint, and it does not require stowing procedures or additional storage space. This is possible because VMEM are foreseen to function as wall-mounted devices; its rail is mounted to a structurally stable wall beam or established support structure thereby demonstrating a zero footprint.

[0018] In addition, it can be foreseen that alternate embodiments of a VMEM will have a carriage with the capability to function directly on an apt structure or frame (i.e. off-rail). It can also be foreseen that alternate embodiments of a VMEM can be fitted with kinetic power generating devices and/or pumping systems. The corresponding motion of a physically propelled carriage can be utilized to power small electrics, recharge batteries, pressurize systems, and/or pump fluids thereby performing a wide array of activity, work, and/or services that may suit a multitude of needs beyond exercise.

[0019] A VMEM, with-or-without kinetic anchors, is worthwhile. It optimizes functional efficiency; it's safe; it conserves time; it has a zero footprint . . . worthwhile features that can make it a welcomed addition to the current catalog of multifunctional exercise machines.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 is a front view of an embodiment showing a relevant section of a multifunctional exercise machine whereby a carriage is physically propelled along a wall-mounted rail situated in the vertical plane (VMEM).

[0021] FIG. 2 is a side view of an embodiment showing a relevant section of a VMEM.

[0022] FIG. 3 is a top view of an embodiment of the VMEM.

TABLE-US-00001 REFERENCE NUMBERS OF THE DRAWINGS 10 seat pad assembly 12 handles 14 seat pad attachments 16 chassis 18 wheels 20 rail 22 kinetic anchors male 24 kinetic anchors female 26 resistance knob 28 locking mechanism 30 static anchors male 32 static anchors female 34 wall board 36 wall beam 38 pass-through pole accessory 40 wheel tracks 42 carriage 44 VMEM 46 carriage-stop bumper 48 locking hole

DETAILED DESCRIPTION OF THE INVENTION

[0023] The following descriptions are representative of the embodiment set forth in this application.

[0024] This embodiment is of a multifunctional exercise machine whereby a carriage is physically propelled along a wall-mounted rail situated in the vertical plane (VMEM).

[0025] FIG. 1 shows a front view of an embodiment of a relevant section of a VMEM 44. A VMEM 44 is a combination of a physically propelled carriage 42, of which its primary structure comprises a seat pad assembly 10 of predetermined dimensions and characteristics attached to an

appropriately supportive chassis **16** of predetermined dimensions with wheels **18** (FIG. 2) of a predetermined size that are in a functional accordance with wheel tracks **40** of a predetermined dimension. Wheel tracks **40** are situated along a vertically oriented rail **20** of predetermined dimensions that is mounted on top of wall board **34** (FIG. 2) and secured to an underlying wall beam **36** (FIG. 2). Shown on the carriage **42** are kinetic anchors male **22** with predetermined characteristics and kinetic anchors female **24** with predetermined characteristics along with a pass-through pole accessory **38** of a predetermined size. Along the rail **20** there are static anchors male **30** with predetermined characteristics. The carriage **42** is physically propelled along the vertically oriented rail **20** to perform exercise. Kinetic anchors male **22** are anchoring points for attachments and other accessories that can be utilized in conjunction with the movement of the carriage **42** along the rail **20** to perform upper extremity and torso activities while simultaneously training lower extremity and torso muscle groups along with stabilizers while in the vertical plane. Static anchors male **30** are anchoring points for attachments and other accessories that can be utilized in conjunction with kinetic anchors male **22** or kinetic anchors female **24** to create resistance to the physically propelled movement of the carriage **42** along the rail **20** or independently to perform upper extremity and torso activities while simultaneously training lower extremity and torso muscle groups along with stabilizers while physically propelling the carriage **42** along the rail **20** in the vertical plane.

[0026] FIG. 2 shows a side view of an embodiment of a relevant section of a VMEM **44**. The VMEM **44** is a combination of a physically propelled carriage **42**, of which its primary structure comprises a seat pad assembly **10** of predetermined dimensions and characteristics attached to an appropriately supportive chassis **16** of predetermined dimensions with wheels **18** of a predetermined size that are in a functional accordance with wheel tracks **40** of a predetermined dimension. Wheel tracks **40** are situated along a vertically oriented rail **20** of predetermined dimensions that is mounted on top of wall board **34** and secured to an underlying wall beam **36**. The seat pad assembly **10** is connected to an appropriately supportive chassis **16** by seat pad attachments **14** of predetermined dimensions and characteristics. Affixed to the carriage **42** are kinetic anchors male **22**, kinetic anchors female **24**, handles **12**, a resistance knob **26**, and a locking mechanism **28**, all of which have predetermined dimensions and characteristics. Along the rail **20** there are static anchors male **30** and static anchors female **32**, both having predetermined dimensions and characteristics. Situated inside the wheel track **40** at the inferior end of the rail **20** is the carriage-stop bumper **46** of predetermined dimensions and characteristics. The carriage **42** is physically propelled along the vertically oriented rail **20** to perform exercise. Kinetic anchors male **22** and kinetic anchors female **24** are respectively anchoring points and openings for attachments and other accessories that can be utilized in conjunction with the movement of the physically propelled carriage **42** along the rail **20** to perform upper extremity and torso activities while simultaneously training lower extremity and torso muscle groups along with stabilizers while in the vertical plane. The carriage **42** can be locked at various locations along the rail **20** utilizing the locking mechanism **28** in conjunction with locking holes **48** of a predetermined size. Static anchors male **30** and static anchors female **32** are anchoring points for attachments and other accessories that can be utilized in conjunction with kinetic anchors male **22** or kinetic anchors female **24** to create resistance to the movement of the carriage **42** along the rail **20** or independently to perform upper extremity and torso activities while simultaneously training lower extremity and torso muscle groups along with stabilizers while physically propelling the carriage **42** along the rail **20** in the vertical plane.

[0027] FIG. 3 shows a top view of an embodiment of a VMEM **44**. The VMEM **44** is a combination of a physically propelled carriage **42** of which its primary structure comprises a seat pad assembly **10** of predetermined dimensions and characteristics attached to an appropriately supportive chassis **16** of predetermined dimensions with wheels **18** of a predetermined size that are in a functional accordance with wheel tracks **40** of a predetermined dimension. Wheel tracks **40** are

situated along a vertically oriented rail **20** of predetermined dimensions that is mounted on top of wall board **34** and secured to an underlying wall beam **36**. Shown mounted to the rail **20** there is a static anchor male **30** with predetermined characteristics. The seat pad assembly **10** is connected to an appropriately supportive chassis **16** by seat pad attachments **14** with predetermined dimensions and characteristics. Affixed to the carriage **42** are kinetic anchors male **22**, kinetic anchors female **24** with a pass-through pole accessory **38**, handles **12**, a resistance knob **26**, and a locking mechanism **28**, all of which have predetermined dimensions and characteristics. The carriage **42** is physically propelled along the rail **20** to perform exercise in the vertical plane. Kinetic anchors male **22** and kinetic anchors female **24** are respectively anchoring points and openings for attachments and other accessories that can be utilized in conjunction with the physically propelled movement of the carriage **42** along the rail **20** to perform upper extremity and torso activities while simultaneously training lower extremity and torso muscle groups along with stabilizers while in the vertical plane. Static anchors male **30** and static anchors female **32** (FIG. 2) both with predetermined dimensions and characteristics are anchoring points for attachments and other accessories that can be utilized in conjunction with kinetic anchors male **22** or kinetic anchors female **24** to create resistance to the movement of the carriage **42** along the rail **20** or independently to perform upper extremity and torso activities while simultaneously training lower extremity and torso muscle groups along with stabilizers while physically propelling the carriage **42** along the rail **20** in the vertical plane.

Claims

1. A multifunctional exercise machine whereby a carriage is physically propelled along a wall-mounted rail situated in the vertical plane.
 2. The multifunctional exercise machine whereby a carriage is physically propelled along a wall-mounted rail situated in the vertical plane as set forth in claim 1, comprising: a. anchoring points for attachments and other accessories situated on said carriage that move in synchrony with said carriage, b. openings for said attachments and said other accessories on said carriage that move in synchrony with said carriage.
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